

Home Credit Default Risk

Data Cleaning & Preprocessing — Notebook Analysis Report

Source reviewed: the uploaded `data_cleaning_and_preprocessing.ipynb` and its executed outputs. This report separates observed facts from recommended actions. An unusual value is not treated as an error without evidence.

1. Executive Summary

Finding	What the notebook shows	Recommended interpretation
Dataset size	307,511 rows × 1,076 columns; ~2.5 GB reported memory usage	Large merged tabular dataset. Avoid unnecessary full copies.
Duplicates	0 duplicate rows detected.	No duplicate-row removal is currently required.
Target	TARGET mean ≈ 0.0807, about 8.07% defaults.	Strong class imbalance. Handle during modeling, not by deleting majority rows.
Missingness	335 columns >60% missing; 35 at 40–60%; 14 at 20–40%; 627 at 5–20%; 10 at <5%; 55 with no missing	Missingness is a major preprocessing issue, especially in merged historical/aggregate features.
DAYS_BIRTH	No missing values; range –25,229 to –7,489 days.	Negative values are intentional relative-day encoding, not invalid ages.
DAYS_EMPLOYED	Maximum 365,243 while ordinary values are negative.	365,243 is a special/anomalous sentinel, not a real employment duration.
Financial extremes	AMT_INCOME_TOTAL max 117,000,000 vs median 147,150; AMT_CREDIT max 1,050,000; AMT_ANNUITY max 258,025.5.	Please AMT_CREDIT investigate outliers; AMT_INCOME_TOTAL and AMT_ANNUITY

2. What the Notebook Has Established

- **Data structure:** 1,076 columns: 1,019 float, 41 integer and 16 string columns.
- **Duplicates:** 0 duplicate rows detected.
- **Missingness:** per-column counts and percentages were calculated and grouped.
- **Types:** numerical and categorical columns were separated for later preprocessing.
- **Special values:** negative numerical values and selected financial variables were inspected.
- **Outliers:** box plots were created for AMT_INCOME_TOTAL, AMT_CREDIT and AMT_ANNUITY; no deletion has been justified yet.

3. Missing-Value Findings

The notebook reports this distribution across the 1,076 columns:

Missingness band	Number of columns
>60%	335
40–60%	35
20–40%	14
5–20%	627
<5%	10
No missing values	55

Examples include several `PREV_RATE_INTEREST_*` features with 98.50% missingness and many `PREV_CC_*` aggregate features around 82.8%. Original application variables also have substantial missingness: `OWN_CAR_AGE` has 202,929 missing values (≈66.0%), `OCCUPATION_TYPE` has 96,391 (≈31.3%), and `EXT_SOURCE_1` has 173,378 (≈56.4%).

Recommended action: Do not fill all NaNs with zero. First distinguish 'no historical record' from a genuinely missing measurement. High-missingness features should be candidates for pruning/testing, not automatically deleted.

4. Why DAYS_BIRTH Is Negative

DAYS_BIRTH is not a calendar date stored as a negative number. It is a relative number of days between the application reference date and the applicant's birth date. Birth occurred before the application date, so the interval is negative.

For example, -9,461 days corresponds to about 25.9 years. The notebook's range (-25,229 to -7,489) corresponds to roughly 69.1 to 20.5 years. The median of -15,750 days is roughly 43.2 years.

Recommended action: Do not delete these negatives. During feature engineering create an interpretable **AGE_YEARS = -DAYS_BIRTH / 365.25** feature. The original DAYS_BIRTH can then be retained or excluded depending on the final pipeline.

5. Other Negative Values and Sentinel Values

DAYS_EMPLOYED, DAYS_REGISTRATION and DAYS_ID_PUBLISH are also relative-day fields, so negative values can be valid. Therefore, a rule such as 'delete every negative numerical value' would be wrong.

DAYS_EMPLOYED = 365243 is different: it is a special sentinel/anomalous code in this dataset, not a plausible employment duration. Recommended treatment is to convert this code to missing and optionally create a flag such as DAYS_EMPLOYED_ANOMALY before deriving employment-duration features.

6. Outlier Findings

The notebook creates box plots for three monetary variables. A box plot flags points beyond its whiskers as statistical outliers. That means 'unusual relative to the distribution', not 'wrong'.

Feature	Median	Maximum	Interpretation
AMT_INCOME_TOTAL	147,150	117,000,000	Very right-skewed; extreme high-income observations exist.
AMT_CREDIT	513,531	4,050,000	Large spread; high loan values can be legitimate.
AMT_ANNUITY	24,903	258,025.5	Extreme values exist; investigate before treatment.

Recommended action: Do not delete outlier rows now. Investigate percentiles, skewness, counts and business meaning. For heavily right-skewed monetary variables, log transformation or models robust to skew may be appropriate. Capping/winsorization should be tested rather than assumed.

7. Columns That Are Candidates for Removal

Candidate type	Evidence	Action
>95% missing	Several PREV_RATE_INTEREST_* features at 99.50% missing	Strong candidate for removal; validate predictive value and meaning of missing
80–95% missing	Many PREV_CC_* aggregates around 82.8% missing	Strong candidate for pruning/model-based selection; do not delete automatically
High missingness in original features	OWN_CAR_AGE ≈66%; EXT_SOURCE_1 ≈56%	Investigate meaning and predictive value; missingness indicators may help.
Near-constant columns	FLAG_MOBIL is essentially always 1	Strong candidate for removal after exact uniqueness/variance checks.
Identifier	SK_ID_CURR	Keep for joins/identification; normally exclude from model features.

8. Recommended Cleaning Plan

- 1. Keep the raw train_merged.csv unchanged.
- 2. No duplicate removal is needed based on the current 0-duplicate result.
- 3. Handle DAYS_EMPLOYED = 365243 explicitly.
- 4. Convert relative-day fields into interpretable features such as age and employment duration.

- **5.** Audit missingness by feature family: application vs historical/aggregate features.
- **6.** Consider missingness indicators when missing means absence of a record.
- **7.** Identify constant/near-constant features before deletion.
- **8.** Investigate outliers; do not mass-delete them.
- **9.** Keep encoding and feature engineering as separate later stages.
- **10.** Record before/after shape, missingness and special-value counts after every major transformation.

9. What NOT to Do

- Do not delete all rows containing outliers.
- Do not turn every negative value into a positive value.
- Do not fill every NaN with 0.
- Do not drop every column with >50% missing automatically.
- Do not balance TARGET by deleting the majority class.
- Do not fit imputers/scalers on validation/test data; fit them using training data only.

10. Current Project Position

Roadmap stage	Status
1. Understand problem	Completed
2. Understand dataset	Completed
3. Data cleaning	Investigation completed; treatment pending
4. Missing-value analysis & treatment	Analysis completed; treatment pending
5. Outlier analysis & treatment	Initial analysis started; treatment pending
6. EDA	Mostly completed in previous notebooks
7. Categorical encoding	Not started
8–18	Not started

Bottom line: The data is not 'bad' because DAYS_BIRTH is negative or because many values are missing. Those are characteristics of the dataset that need correct interpretation. The main challenge is the very wide merged feature set and heterogeneous missingness. The next step should be feature-family-aware cleaning, not blanket deletion or imputation.